

Data-Driven Decision Support System For Marketing Campaign Optimisation

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Abstract. One of the challenges faced by e-commerce sellers is assessing advertising performance across a number of online platforms. This study presents a mobile-based Decision Support System (DSS) for advertising performance evaluation and platform comparison. The system was developed using Android Studio, Firebase Authentication, and the Simple Additive Weighting (SAW) method. It enables users to upload CSV files, automatically calculates the most important Key Performance Indicators (KPIs) such as Return on Investment (ROI), Return on Advertising Spend (ROAS), Click-Through Rate (CTR), Conversion Rate (CVR) and Cost Per Click (CPC), and produces platform rankings based on weights. The system functions and calculation of KPIs were tested to ensure accuracy. The results indicate that the proposed DSS can effectively evaluate advertising performance and guide e-commerce sellers to choose the most suitable advertising platform. The system offers a practical and efficient decision support to enhance the advertising strategy and the evaluation of the advertising campaigns.

INTRODUCTION

With its potential for reaching a broad audience and driving sales, digital advertising has become a vital tool for e-commerce businesses. Popular e-commerce platforms like Shopee, Lazada and TikTok offer different advertising platforms with different dashboards and formats. This makes it challenging for sellers to measure the performance of their advertising across various platforms and to find the best platform for their products. Judging advertising by personal experience alone can result in poor marketing decisions and wasted budget.

A decision support system can help e-commerce sellers evaluate advertising performance more efficiently. The current study developed a mobile-based DSS that can analyse uploaded advertising data in CSV format and present the results on Key Performance Indicator (KPI) tables, data visualisations, and platform ranking. Five KPIs are used to measure the performance of advertising: ROI, ROAS, CTR, CVR and CPC. The Simple Additive Weighting (SAW) method is used to rank the advertising platforms based on predefined KPI weights.

The primary goal of this project is to offer a user-friendly platform that empowers e-commerce merchants to make better advertising choices by enabling them to compare the performance of advertising on Shopee, Lazada, and TikTok. The proposed system enhances the efficiency of the decision making process, particularly for those new sellers who have less experience in advertising, by converting advertising data into useful information. The system is mainly used to analyse uploaded CSV files and provides results for use as decision support; it does not support real-time advertisement monitoring or automatic campaign management.

The remainder of this paper is organised as follows. A review of the related studies about decision support systems and advertising performance evaluation is discussed in Section 2. The proposed methodology and system development are described in Section 3. Section 4 gives the implementation and experimental results, and Section 5 concludes the paper and discusses future work.

RELATED WORK

Decision Support Systems (DSS) are widely used for complex and semi-structured decision-making by integrating data analysis with decision models. DSS has been successfully applied in a number of applications including employee performance evaluation, marketing strategy selection and resource allocation to enhance the quality of decisions and to decrease the role of subjective judgement¹⁻⁶.

The Simple Additive Weighting (SAW) method is widely used among different multi-criteria decision-making (MCDM) methods due to its simplicity, efficiency in calculations and interpretability. SAW normalises the criterion values, weights each criterion, and determines a weighted score for each alternative to rank them. The advantage of the SAW method over other more complex methods, such as Analytic Network Process (ANP), Fuzzy methods, and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is that it is less computationally complex and allows for transparent ranking results, which makes it suitable for practical decision support applications^{1,3,6}.

In e-commerce advertising, there are several Key Performance Indicators (KPIs) that are used to measure a campaign's performance, such as Return on Investment (ROI), Return on Advertising Spend (ROAS), Click-Through Rate (CTR), Conversion Rate (CVR) and Cost Per Click (CPC)⁷⁻⁹. These indicators are measuring various components of the advertising effectiveness, so the choice of the most appropriate advertising platform turns into a multi-criteria decision making. So, combining the KPI evaluation with the SAW method is a practical solution that enables the comparison of the performance of advertising and helps in data-driven decisions. The calculated KPI values are used as the evaluation criteria in the SAW method to prioritise advertising platforms and support decision-making for e-commerce sellers.

METHODOLOGY

3.1 System Framework

To help e-commerce sellers analyse performance in advertising campaigns and across various platforms, the proposed Decision Support System (DSS) was created. The exported data from Shopee, Lazada and TikTok in CSV format is uploaded into the system, and then the uploaded data is processed, such as data validation, calculation of Key Performance Indicators (KPIs), Simple Additive Weighting (SAW) evaluation and visualisation of the results. The final product is a series of KPI comparison tables, charts and platform rankings that help users select the most suitable advertising platform

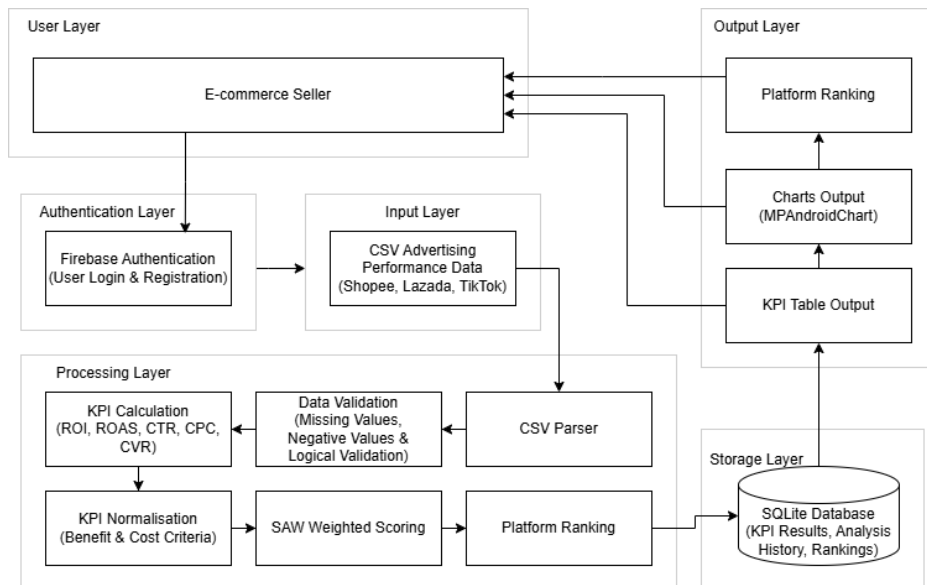


Figure 1. System Architecture Diagram

Figure 1 shows the overall system architecture. The system has six layers: User Layer, Authentication Layer, Input Layer, Processing Layer, Storage Layer and Output Layer. The user registration and login are secured by Firebase Authentication, and the KPI analysis results and ranking history are stored in SQLite. In the process, the uploaded CSV data is analysed, validated, transformed into KPIs, normalised and evaluated with the SAW method and finally presented to users.

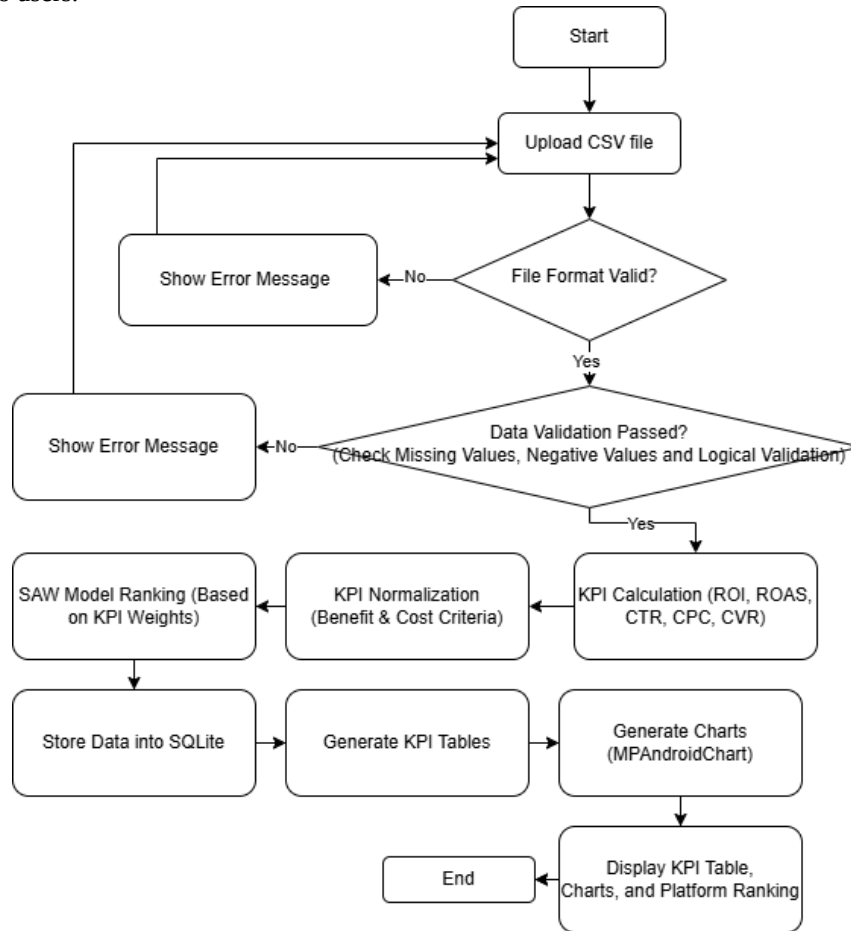


Figure 2. System Flowchart

Figure 2 shows the system workflow. Once the advertising data is uploaded by the users, the system validates the uploaded data, calculates the KPIs, normalises them using SAW and weighted scores, stores the analysis results in the SQLite database, and finally produces a set of tables of KPIs, visualisation charts and platform rankings.

3.2 Data Processing

The proposed DSS imports advertising performance datasets from Shopee, Lazada, and TikTok into the proposed DSS in CSV format. Every uploaded dataset includes advertising data, including impressions, clicks, orders, advertising spend and revenue. Once the file is uploaded, the system determines which advertising platform is being used and collects the necessary fields for analysis.

The data uploaded before calculating KPIs is validated to ensure the quality of the data. The system checks if all the required fields are there and if they have any numeric values and do not have any negative values. Logical validation is also carried out to ensure that data is consistent. Specifically, impressions and ad spend should be positive, and the number of ad clicks should not be greater than the number of impressions. Invalid data are rejected automatically, and an error message is presented to warn the user from entering invalid data, thus avoiding incorrect calculations and unreliable decision-making results.

3.3 KPI Calculation

The system calculates five Key Performance Indicators (KPIs) used to measure the effectiveness of ads. The above-mentioned KPIs are the criteria for making the decision in the proposed DSS.

Table 1. KPI Formulas and Criterion Types

KPI	Formula	Criterion Type
Return on Investment (ROI)	(Revenue – Spend) / Spend	Benefit
Return on Advertising Spend (ROAS)	Revenue / Spend	Benefit
Click-Through Rate (CTR)	Clicks / Impressions	Benefit
Conversion Rate (CVR)	Orders / Clicks	Benefit
Cost Per Click (CPC)	Spend / Clicks	Cost

Table 1 shows the KPI criterion, formula and criterion type. ROI measures the profitability of advertising, and ROAS measures the revenue generated from advertising expenditure. CTR measures user engagement with advertisements, CVR measures the proportion of users who complete a purchase after clicking an advertisement, and CPC is the cost of a click. The CPC is a cost criterion; other criteria are considered benefit criteria for the SAW evaluation process. The weights were determined based on the relative importance of each KPI in evaluating advertising performance.

3.4 SAW Decision Model

Multiple KPIs are used to rank advertising platforms with the Simple Additive Weighting (SAW) method. SAW was selected because of its simplicity, computational efficiency, and ease of interpretation. In this study, the decision alternatives are Shopee, Lazada, and TikTok. These are five KPI criteria that have predefined weights, which are given in Table 2.

The SAW evaluation consists of three main steps. Firstly, advertising data is uploaded, and KPI values are calculated based on this data. Second, all the criteria are normalised to make them comparable with the following equations:

Table 2. Criteria Weights for SAW Evaluation

Criterion	Weight	Attribute
ROI	0.40	Benefit
ROAS	0.30	Benefit
CTR	0.15	Benefit
CVR	0.10	Benefit
CPC	0.05	Cost

Table 2 shows each criterion with its weight and attributes. The weight and attribute are used in SAW calculations; below are formulas for SAW and normalisation calculations.

X_i = Original value of criterion i

Y_i = Normalised value of criterion i

$Max(X)$ = Highest value among all alternatives for the criterion

$Min(X)$ = Lowest value among all alternatives for the criterion

Formula for benefit criterion:

$$Y_i = \frac{X_i}{Max(X)} \quad (1)$$

Formula for cost criterion:

$$Y_i = \frac{Min(X)}{X_i} \quad (2)$$

Finally, the overall SAW score for each platform is calculated using the weighted sum equation:

F = Final SAW score of an alternative (platform)

W_i = Weight assigned to criterion i

Y_i = Normalised value of criterion i

i = Criterion index

n = Total number of criteria

General Formula:

$$F = \sum_{i=1}^n W_i Y_i \quad (3)$$

Applied Formula:

$$F = (W_{ROI} \times Y_{ROI}) + (W_{ROAS} \times Y_{ROAS}) + (W_{CTR} \times Y_{CTR}) + (W_{CVR} \times Y_{CVR}) + (W_{CPC} \times Y_{CPC}) \quad (4)$$

The final SAW score is calculated by multiplying each normalised criterion value by its corresponding weight and summing the weighted values. The platform with the highest SAW score is considered the best alternative.

3.5 System Implementation

The development of the DSS application was done in the Java programming language using Android Studio. Firebase Authentication was integrated to manage secure user registration, login and password recovery, while SQLite was used as the local database to store advertising data uploaded, KPI results, analysis history and SAW rankings.

The advertising datasets are uploaded by the users using the mobile application in CSV format. In data validation, the system automatically calculates values of the KPIs, performs the SAW evaluation and produces comparison results. MPAndroidChart is used to visualise KPI performance in charts, and comparison tables and ranking results give the user clear information about the performance of advertising on other e-commerce platforms. Users can analyse the effectiveness of ads efficiently with mobile implementation, automated KPI analysis and SAW ranking without manual calculation.

RESULTS AND DISCUSSION

4.1 System Implementation

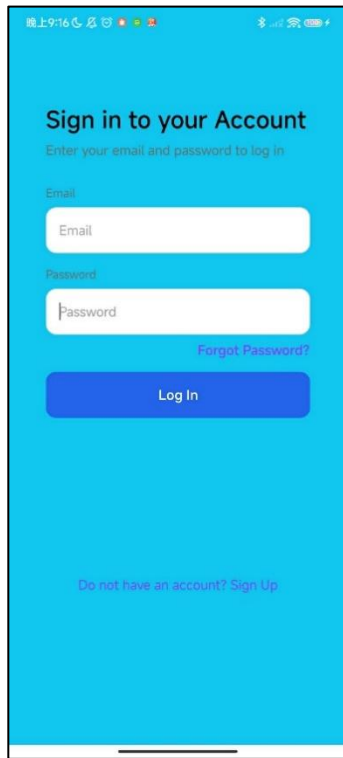


Figure 2. Login Page



Figure 3. Upload CSV Page

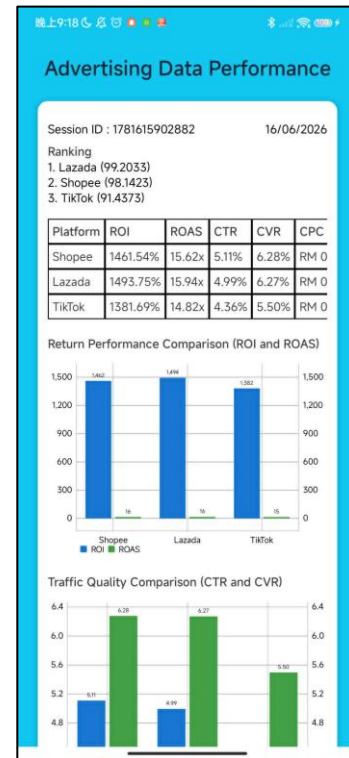


Figure 4. KPI Results, Platform Ranking and Visualisation Page

The proposed DSS was successfully developed as an Android mobile application which helps e-commerce sellers analyse the performance of advertising on various platforms. It uses Firebase Authentication to handle user management securely, SQLite to store the data locally, and MPAndroidChart to visualise KPI results.

The developed system consists of the major functions as shown in Figures 3, 4, and 5. The users log in to the application using their registered accounts. The user needs to upload advertising data in CSV format after authentication for advertising on Shopee, Lazada, and TikTok. The uploaded data are automatically validated, the KPIs are calculated, the SAW evaluation is performed, and comparison results are generated. The system performs the analysis process and generates KPI comparison tables, visualisation charts, and platform rankings. Furthermore, the results of the analysis performed are saved in SQLite, and users can see previously done advertising analyses in the history module.

4.2 Functional Testing

The developed DSS has undergone functional testing to ensure that all pertinent functions are working correctly. Testing was conducted on user authentication, CSV upload, calculation of KPIs, SAW ranking, data visualisation and history management. All test cases were passed successfully, which means that the implemented functions meet the requirements of the system.

Table 3. Important Features Testing

Function	Expected Result	Status
User Authentication	User can register and log in successfully	Pass
CSV Upload	Valid CSV files are accepted for analysis	Pass
KPI Calculation	ROI, ROAS, CTR, CVR and CPC are calculated correctly	Pass
SAW Ranking	Platforms are ranked according to weighted KPI scores	Pass
Data Visualisation	KPI tables and charts are displayed correctly	Pass
History Management	Analysis results are stored and retrieved successfully	Pass

Table 3 shows testing results that demonstrate that the developed application performs reliably and all essential functions operate as expected.

4.3 KPI and SAW Evaluation

The KPI values obtained from the system were also compared with manual calculations, using the same advertising dataset, to check the correctness of the proposed DSS. The results obtained from the comparison showed that the formulas used in the KPIs implemented were accurate.

Table 4. Shopee Campaign KPI and SAW Evaluation Results

KPI	Manual Calculation	System Result
ROI	3.533	353.33%
ROAS	4.533	4.53
CTR	0.043	0.043
CVR	0.033	0.034
CPC	0.536	RM 0.54

Table 4 shows that the differences between the manual calculations and the system results are minor and are mainly due to decimal rounding and percentage formatting. In summary, the system results obtained KPI values were correlated with manual calculations, and the results are correct for the KPI formulas implemented.

The KPI values calculated were then fed into the SAW to rank the advertising performance of various e-commerce platforms. The final SAW evaluation results are shown in Table 5.

Table 5. Platform Ranking Based on SAW Scores

Platform	SAW Score	Ranking
Lazada	0.9544	1
Shopee	0.9315	2
TikTok	0.9182	3

Table 5 shows that Shopee achieved the highest SAW score among all evaluated platforms. Therefore, Shopee is identified as the best advertising platform based on the selected KPI criteria and their assigned weights.

4.4 Discussion

The experimental findings show that the proposed DSS can be successfully implemented in a mobile application for the assessment of advertising performance by combining the analysis of KPIs and the SAW decision-making process. The developed system can automatically validate uploaded datasets, compute various KPIs, rank the

platforms, and display the results in tables and visualisations, which can help e-commerce sellers save time to understand the advertising performance and reduce calculation errors when compared to manual analysis.

Combining KPI evaluation and SAW allows users to make an objective comparison of the advertisement performance on several different e-commerce platforms. The visualisation module also enhances the interpretation of the results by displaying the advertising performance in a clear way. The proposed DSS is useful for small and medium scale e-commerce sellers because of these features.

The overall functional test and evaluation outcomes corroborate that the implemented system is an effective and reliable decision support system for choosing advertising platforms and enhancing decision-making in advertising.

CONCLUSION

This study developed a mobile-based Decision Support System (DSS) that helps e-commerce sellers to analyse and compare ad performance on Shopee, Lazada, and TikTok. The proposed system combines data preparation, validation, calculation of metrics and KPIs, and uses the Simple Additive Weighting (SAW) method to create platform ranking according to several advertising performance indicators. The KPI calculation results and the ranking results in the SAW were found to be correct in both functional testing and manual verification, so they can be used to assist the evaluation and decision making related to advertisements.

The system developed is a viable solution to compare the performance of ads, but some limitations still exist. The current implementation only works with three e-commerce platforms and is based on CSV files that are uploaded manually and not connected to the e-commerce systems in real-time. In addition, the SAW criteria weights are predefined and cannot be customised by users. To add more e-commerce platforms, real-time data collection through API integration, dynamic weight adjustment, and machine learning techniques for predictive advertising recommendations would be beneficial for future projects.

REFERENCES

- 1 M.M. Putri, W. Cholil, S. Afrizal, I.W. Widi Pradnyana, Tjahjanto, and R.M. Bunga Madu, "Decision Support System for Selecting Wedding Venues in Jakarta Using the Simple Additive Weighting (SAW) Method," in 2024 Int. Conf. Inform. Multimed. Cyber Inf. Syst. ICIMCIS, (IEEE, Jakarta, Indonesia, 2024), pp. 662–667.
- 2 Rakshitha, P. S, A. Namjoshi, M. Mane, I.A. Mohammed, and R. Shettigar, "Data Analytics for Decision Support Systems in Techno-Driven Enterprises," in 2025 First Int. Conf. Adv. Comput. Sci. Electr. Electron. Commun. Technol. CE2CT, (IEEE, Bhimtal, Nainital, India, 2025), pp. 1009–1012.
- 3 M.K. Remacle, and Y. Handoko, "Decision Support System Using Simple Additive Weighting with Key Performance Index for Selecting Teacher," in 2024 Int. Conf. Inform. Eng. Sci. Amp Technol. INCITEST, (IEEE, Bandung, Indonesia, 2024), pp. 1–8.
- 4 M.F.N. Azhar, R.H. Laluma, B. Sugiarto, T. Wiharko, D. Kusdian, and D.H.A. Wibowo, "Decision Support Model for Employee Performance Evaluation Using Fuzzy and Weighted Product Methods," in 2025 19th Int. Conf. Telecommun. Syst. Serv. Appl. TSSA, (IEEE, Yogyakarta, Indonesia, 2025), pp. 1–6.
- 5 Nazaruddin, A.A. Purba, M.L. Hamzah, and M. Rizki, "Decision Making for Optimal Marketing Strategy: A Case Study in E-Commerce," in 2022 Int. Conf. Data Anal. Bus. Ind. ICDABI, (IEEE, Sakhr, Bahrain, 2022), pp. 1–5.
- 6 Yuda Irawan, "Decision Support System For Employee Bonus Determination With Web-Based Simple Additive Weighting (SAW) Method In PT. Mayatama Solusindo," J. Appl. Eng. Technol. Sci. JAETS 2(1), 7–13 (2020).
- 7 J. Ghahremani-Nahr, and H. Nozari, "A Survey for Investigating Key Performance Indicators in Digital Marketing," Int. J. Innov. Mark. Elem. 1(1), 1–6 (2021).
- 8 S. Pagampriyal, "The Role of Key Performance Indicators (KPIs) in Social Media Marketing," (2025).
- 9 R. Almestarihi, A.Y.A.B. Ahmad, R.H. Frangieh, I.A. Abu-AlSondos, K.K. Nser, and A. Ziani, "Measuring the ROI of paid advertising campaigns in digital marketing and its effect on business profitability," Uncertain Supply Chain Manag. 12(2), 1275–1284 (2024).